

ASEN 5264 – DMU

Homework #3

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Problem 1:

a)

b) We are given an MDP defined by,

$$S = \mathbb{R}, \quad A = \{0, 0.1, 1\}, \quad \gamma = 1$$

$$T(s' | s, a) = \mathcal{U}([s + a, s + a + 1])$$

$$R(s, a) = -\mathbb{1}(s \leq 1) - a$$

$$U^{\tilde{\pi}}(s) = -2 \cdot \mathbb{1}(s \leq 1)$$

Where $\mathcal{U}([s + a, s + a + 1])$ represents a uniform distribution from $s + a$ to $s + a + 1$, and $\mathbb{1}(s \leq 1) = 1$ if $s \leq 1$ and 0 otherwise. Given $s = 0.8$, we must find the action which maximizes $Q^{\tilde{\pi}}(s, a)$, which is given by

$$Q^{\tilde{\pi}}(s, a) = R(s, a) + \sum_{s'} T(s' | s, a) U^{\tilde{\pi}}(s')$$

Since there are only three actions, we can compute $Q^{\tilde{\pi}}(s = 0.8, a)$ for $\forall a \in A$. Since, S and T are continuous, we also note that the summation will be an integral over all of the possible future state s' . Thus, our expression for $Q^{\tilde{\pi}}(s, a)$ becomes

$$Q^{\tilde{\pi}}(0.8, a) = R(0.8, a) + \int_{-\infty}^{\infty} T(s' | 0.8, a) U^{\tilde{\pi}}(s') ds'$$

We can now compute $Q^{\tilde{\pi}}(0.8, a)$ for all $a \in A$, beginning with $a = 0$. Note that $\mathcal{U}([s + a, s + a + 1]) = 1 \forall a, s$ by definition of the uniform distribution.

a = 0:

$$Q^{\tilde{\pi}}(0.8, 0) = R(0.8, 0) + \int_{-\infty}^{\infty} T(s' | 0.8, 0) U^{\tilde{\pi}}(s') ds'$$

$$Q^{\tilde{\pi}}(0.8, 0) = (-\mathbb{1}(0.8 \leq 1) - 0) + \int_{-\infty}^{\infty} \mathcal{U}([0.8, 1.8]) (-2 \cdot \mathbb{1}(s' \leq 1)) ds'$$

$$Q^{\tilde{\pi}}(0.8, 0) = -1 + \int_{-\infty}^{\infty} (-2 \cdot \mathbb{1}(s' \leq 1)) ds'$$

$$Q^{\tilde{\pi}}(0.8, 0) = -1 + \int_{0.8}^1 -2 ds' = -1 - 2(0.2) = -1.4$$

a = 0.1:

$$Q^{\tilde{\pi}}(0.8, 0.1) = R(0.8, 0.1) + \int_{-\infty}^{\infty} T(s' \mid 0.8, 0.1) U^{\tilde{\pi}}(s') ds'$$

$$Q^{\tilde{\pi}}(0.8, 0.1) = (-\mathbb{1}(0.8 \leq 1) - 0.1) + \int_{-\infty}^{\infty} (-2 \cdot \mathbb{1}(s' \leq 1)) ds'$$

$$Q^{\tilde{\pi}}(0.8, 0.1) = -1.1 + \int_{0.9}^{1.9} (-2 \cdot \mathbb{1}(s' \leq 1)) ds'$$

$$Q^{\tilde{\pi}}(0.8, 0.1) = -1.1 + \int 0.9^1 - 2 ds' = -1.1 - 2(0.1) = -1.3$$

$a = 1$:

$$Q^{\tilde{\pi}}(0.8, 1) = R(0.8, 1) + \int_{-\infty}^{\infty} T(s' \mid 0.8, 1) U^{\tilde{\pi}}(s') ds'$$

$$Q^{\tilde{\pi}}(0.8, 1) = (-\mathbb{1}(0.8 \leq 1) - 1) + \int_{-\infty}^{\infty} (-2 \cdot \mathbb{1}(s' \leq 1)) ds'$$

$$Q^{\tilde{\pi}}(0.8, 1) = -2 + \int_{1.8}^{2.8} (-2 \cdot \mathbb{1}(s' \leq 1)) ds'$$

$$Q^{\tilde{\pi}}(0.8, 1) = -2$$

Thus, the action which minimizes $Q^{\tilde{\pi}}(0.8, a)$ is $\boxed{a = 0.1}$.

Problem 2: